METRIC

FF-B-171/11
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING, ANSI/AFBMA DIMENSION SERIES 02.

The complete requirement for squiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

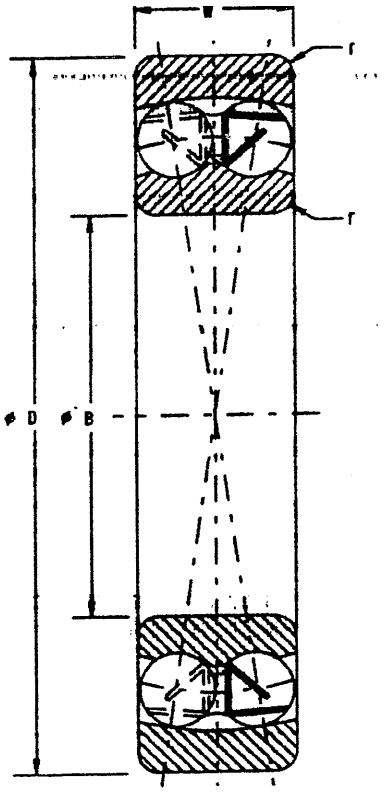


FIGURE 1.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH CYLINDRICAL BORE (CODE C)

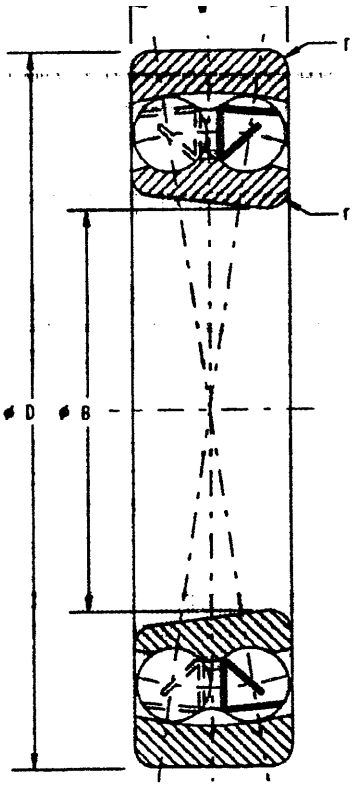


FIGURE 2.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH TAPERED BORE (CODE T)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Design. All self-aligning bearings shell be open. Bores shall be cylindrical or tapered (taper 1:12) and shall be specified in the part number by the codes in table III.

4. Misalignment. Maximum permissible misalignment shall be 4.0 degrees.

5. **Radial** Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

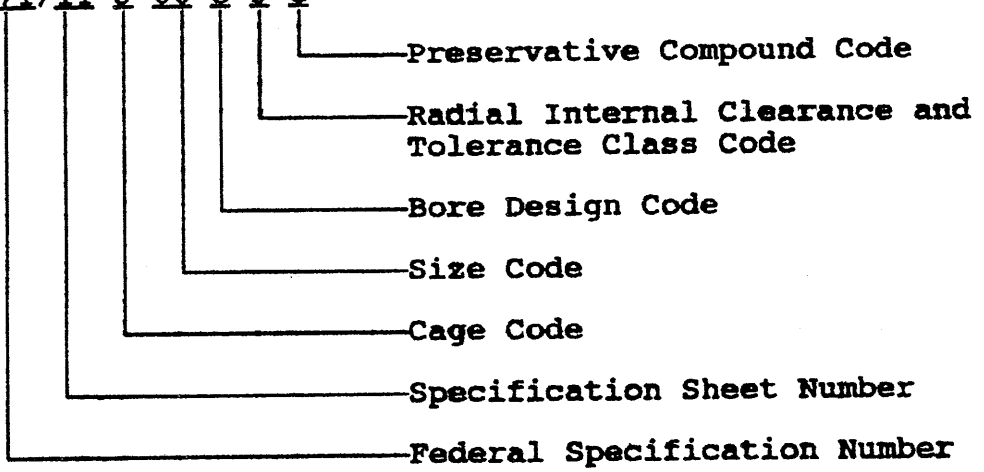
6. Tolerance Class. The tolerance limits for tolerance class ABRC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

7. **Pr**eservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code *C" in the part number.

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8. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, bore design, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/11 J 06 C 5 C



FFB171/11J06C5C indicates - Bearing, cage-pressed steel or molded plastic; bore 30mm; outside diameter 62mm; width 16mm; open (no shields or seals), cylindrical bore; radial internal clearance symbol 2 (less than normal); tolerance class ABEC-5; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	30	9	0.6	15	935
01	12	32	10	0.6	17	990
02	15	35	11	0.6	20	1290
03	17	40	12	0.6	22	1365
04	20	47	14	1.0	26	1715
05	25	52	15	1.0	31	2100
06	30	62	16	1.0	36	2695
07	35	72	17	1.1	42	2750
08	40	80	18	1.1	47	3315
09	45	85	19	1.1	52	2760
10	50	90	20	1.1	57	3915
11	55	100	21	1.5	63.5	4625
12	60	110	22	1.5	68.5	5230
13	65	120	23	1.5	73.5	5360
14	70	125	24	1.5	78.5	6025
15	75	130	25	1.5	83.5	6755
16	80	140	26	2.0	90	6865
17	85	150	28	2.0	95	8535
18	90	160	30	2.0	100	9855
19	95	170	32	2.1	107	11065
20	100	180	34	2.1	112	12015
21	105	190	36	2.1	117	12845
22	110	200	38	2.1	122	14455
24	120	215	40	2.1	132	20600
26	130	230	40	3.0	144	23130
28	140	250	42	3.0	154	26875
30	150	270	45	3.0	164	28860

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

TABLE III. Design Codes.

Code	Design
C	Open (no shields or seals), Cylindrical bore - fig.1
T	Open (no shields or seals), Tapered bore - fig.2

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
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PREPARING
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